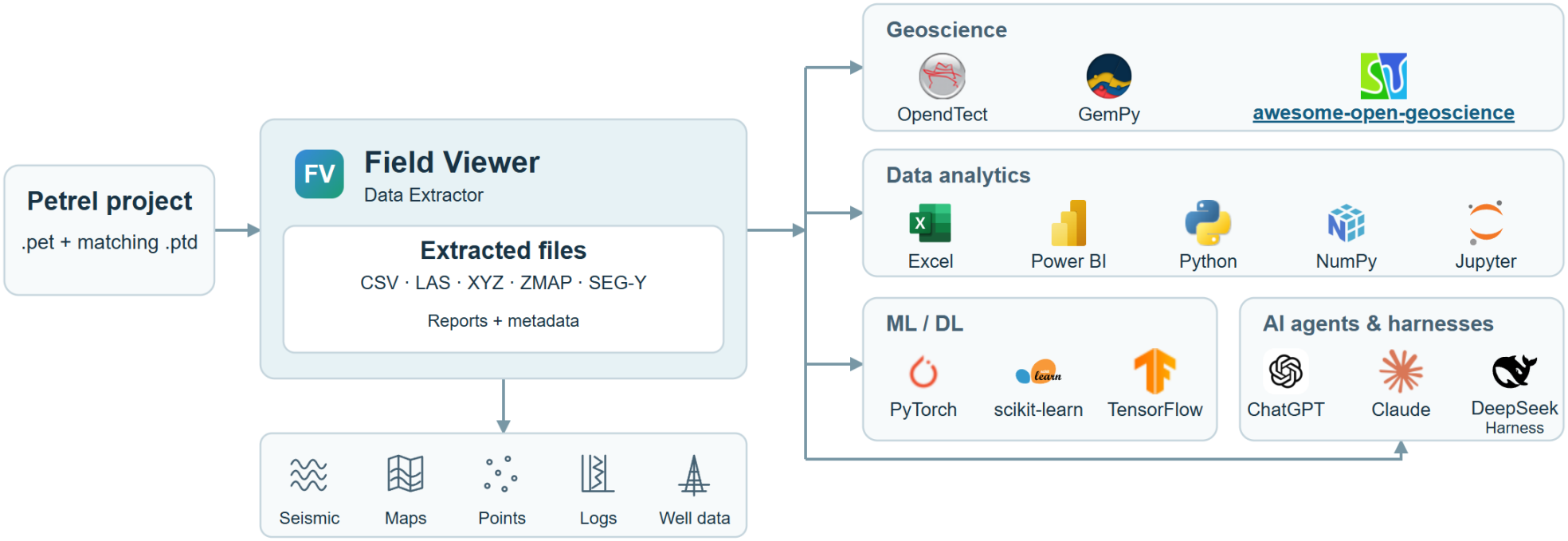
Start Here

Field Viewer Data Extractor

A small tool to help recover supported Petrel data for further work.

By Ahmed Saher Nouh / SaherLabs | Beta 2 (1.0.0-beta.2)



GitHub repository | Download the Windows package | Release notes

The following pages walk through your first extraction.

Start Here

GitHub: Project repository | Download the Windows package | Release notes

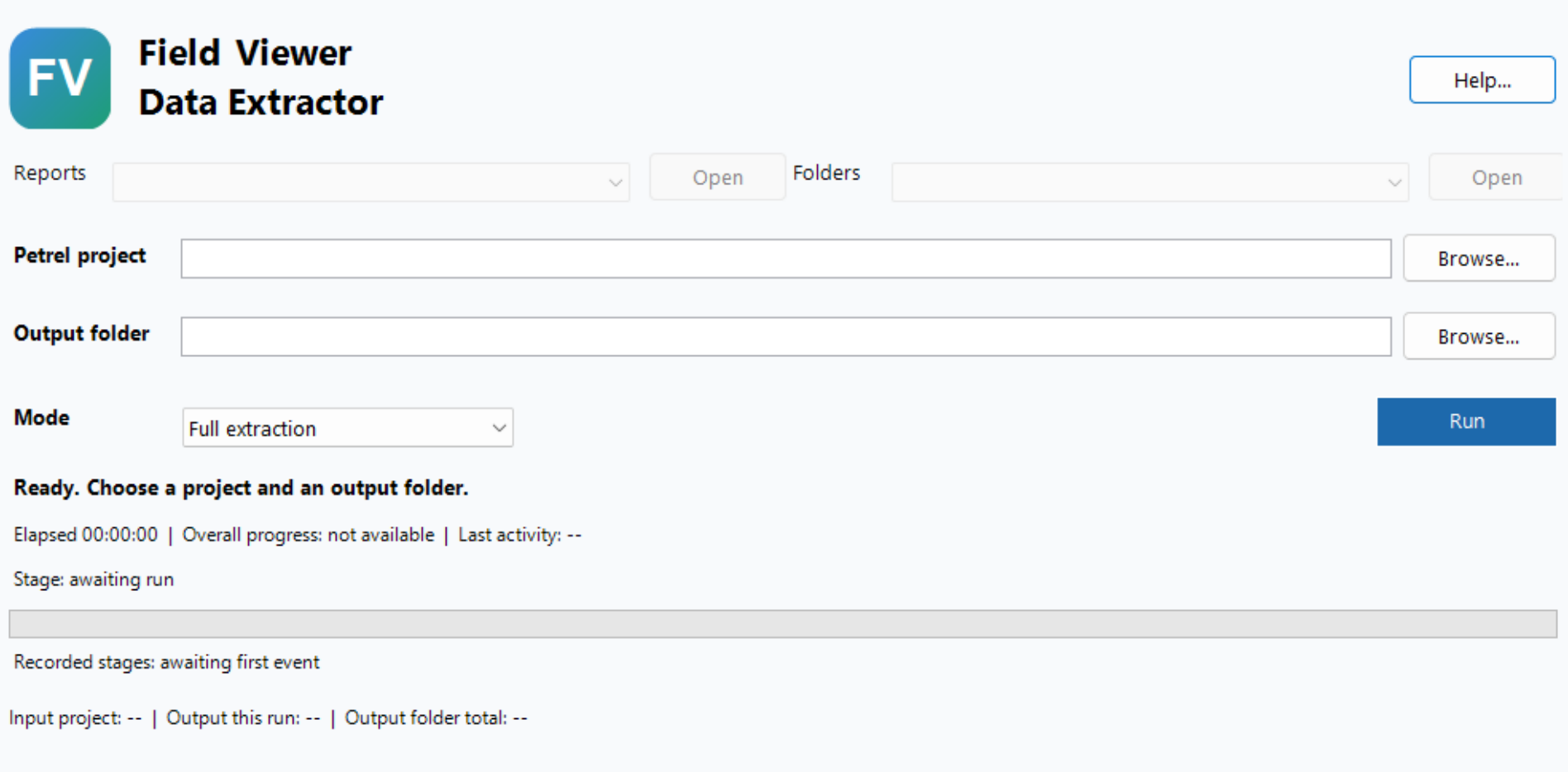
1. Extract the ZIP

You need Windows 10/11 64-bit, a complete Petrel project and enough storage for outputs. Python and Petrel do not need to be installed separately.

1. Download **FieldViewer-Desktop-1.0.0-beta.2-win64.zip** from the GitHub release.
2. In File Explorer, right-click the ZIP and choose **Extract all**. Use a short writable folder, such as **C:\FieldViewer**.
3. Open the extracted folder. Keep every supplied file and folder together.

2. Open Field Viewer Desktop

1. Double-click **FieldViewer_Desktop.bat**. Windows may hide the **.bat** extension.
2. Keep the **desktop** and **FieldViewer** folders beside the launcher. Use the Desktop window for the following steps.
3. The **Headless** download is for command-line use. It has no Desktop window; both editions include the same extraction engine.



Field Viewer Desktop: project, output folder, mode and Run.

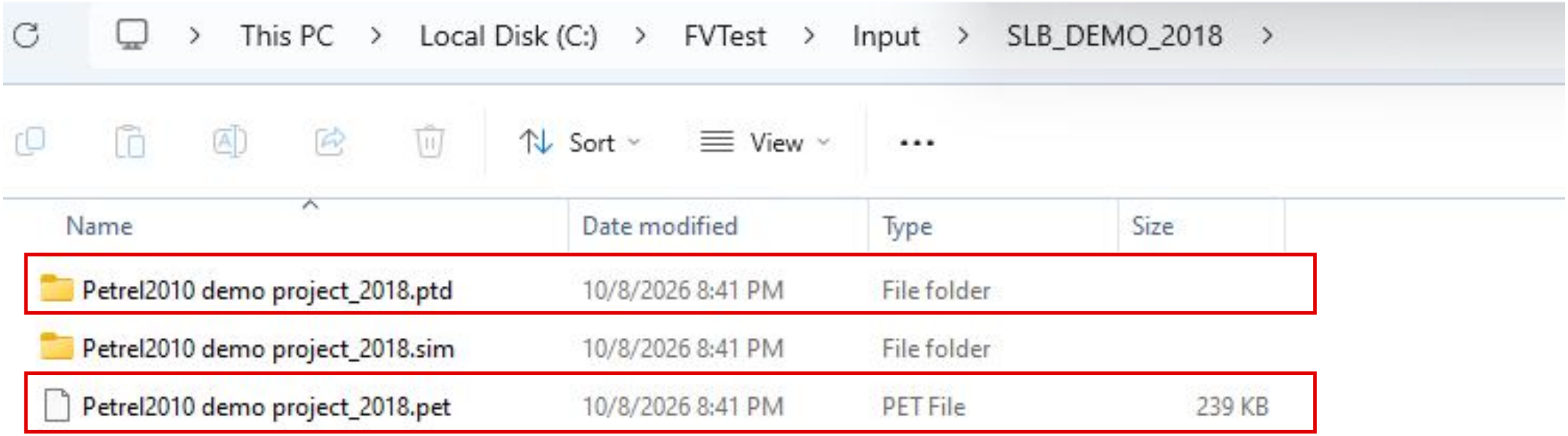
Red outline boxes show what to enter, click or read. Desktop controls and SLB 2018 demo outputs.

GitHub: https://github.com/ahmedsahernouh/petrel-headless-extractor

3. Check your project files

1. Keep the **.pet** file beside its matching **.ptd** folder. Include every subfolder and seismic file.

2. Keep any other supplied project folders with the project. Use a copy and preserve the original files.



The .pet file and matching .ptd folder must stay together.

4. Choose the .pet project

1. Beside **Petrel project**, click **Browse** and select the exact **.pet** file.

2. You can also paste the full path into the **Petrel project** field. Include the filename; its matching **.ptd** folder must be beside it.

Demo example - replace this with your own full project path:

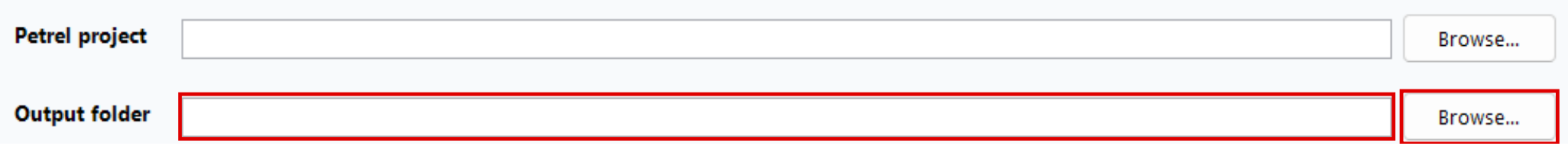
C:\Projects\Demo\Petrel2010 demo project_2018.pet



Choose the .pet project or paste its complete path.

5. Choose an output folder

1. Beside **Output folder**, click **Browse** and select an existing writable folder outside the source project.
2. Prefer a short path such as **C:\FV**. Create that folder in File Explorer first if needed.
3. Each run creates its own **FV_<ID>** folder inside your selected output folder.



Choose a short existing output folder.

6. Select the mode and start

1. Leave **Mode** set to **Full extraction** to recover supported datasets and create reports.
2. Choose **Report only** when you want reports, inventory and previews without retained converted datasets.
3. Click **Run** once. The bundled runtime is prepared and checked automatically.



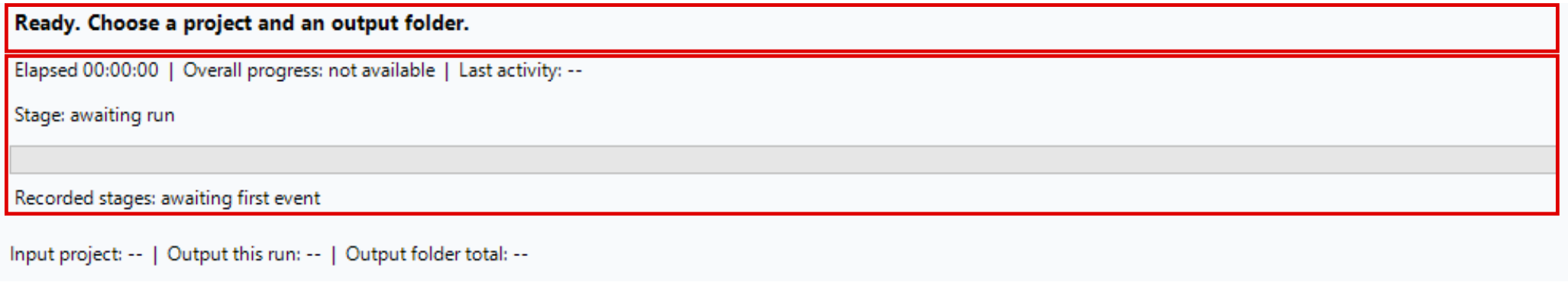
Choose the mode, then click Run once.

7. Wait for the result

1. Follow the status, recorded activity and stage progress. Large projects may take a long time; allow the current run to finish.

2. **Completed with gaps** means outputs are available with stated limitations. Read the report. Exit 10 by itself is not an installation failure.

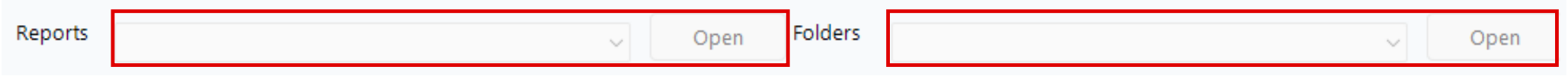
3. Closing the window does not stop extraction. Reopen the same launcher to observe its recorded session. If the run fails, retain the logs and support archive before investigating.



The status and recorded progress appear here while a run is active.

8. Open the report and outputs

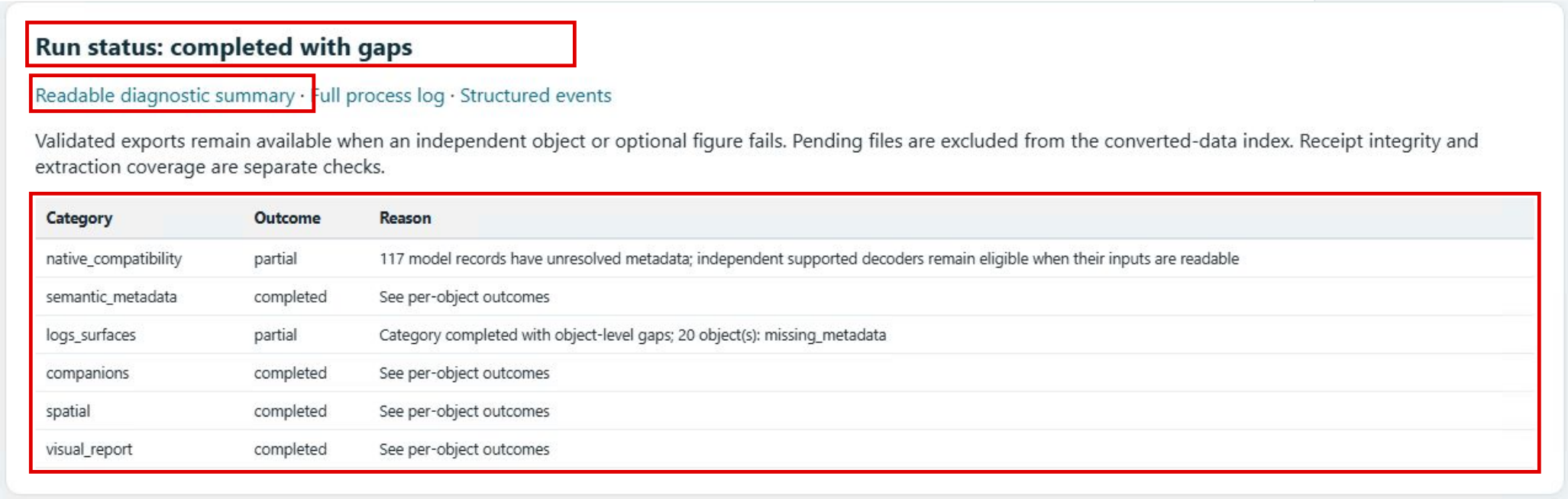
1. When a report is available, select **Main report** in **Reports**, then click the adjacent **Open** button.
2. Select **All exports** in **Folders**, then click its **Open** button to browse recovered data.
3. You can also open the **_REPORT.html** file inside the run folder. Keep it beside its **_EXPORTS** and **_data** folders.



Use the Reports and Folders menus when the destinations become available.

9. Read the status and gaps

1. Read **Run status** and the category outcome table. Check what was recovered and what has missing or unsupported metadata.
2. Click **Readable diagnostic summary** for details when a category is partial. A completed run does not mean every object was decoded.



Run status: completed with gaps

Readable diagnostic summary · Full process log · Structured events

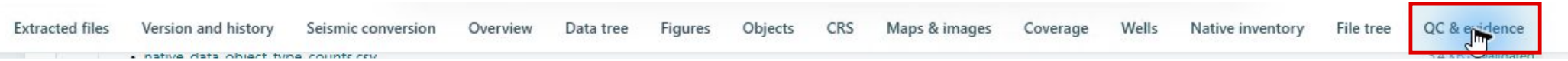
Validated exports remain available when an independent object or optional figure fails. Pending files are excluded from the converted-data index. Receipt integrity and extraction coverage are separate checks.

Category	Outcome	Reason
native_compatibility	partial	117 model records have unresolved metadata; independent supported decoders remain eligible when their inputs are readable
semantic_metadata	completed	See per-object outcomes
logs_surfaces	partial	Category completed with object-level gaps; 20 object(s): missing_metadata
companions	completed	See per-object outcomes
spatial	completed	See per-object outcomes
visual_report	completed	See per-object outcomes

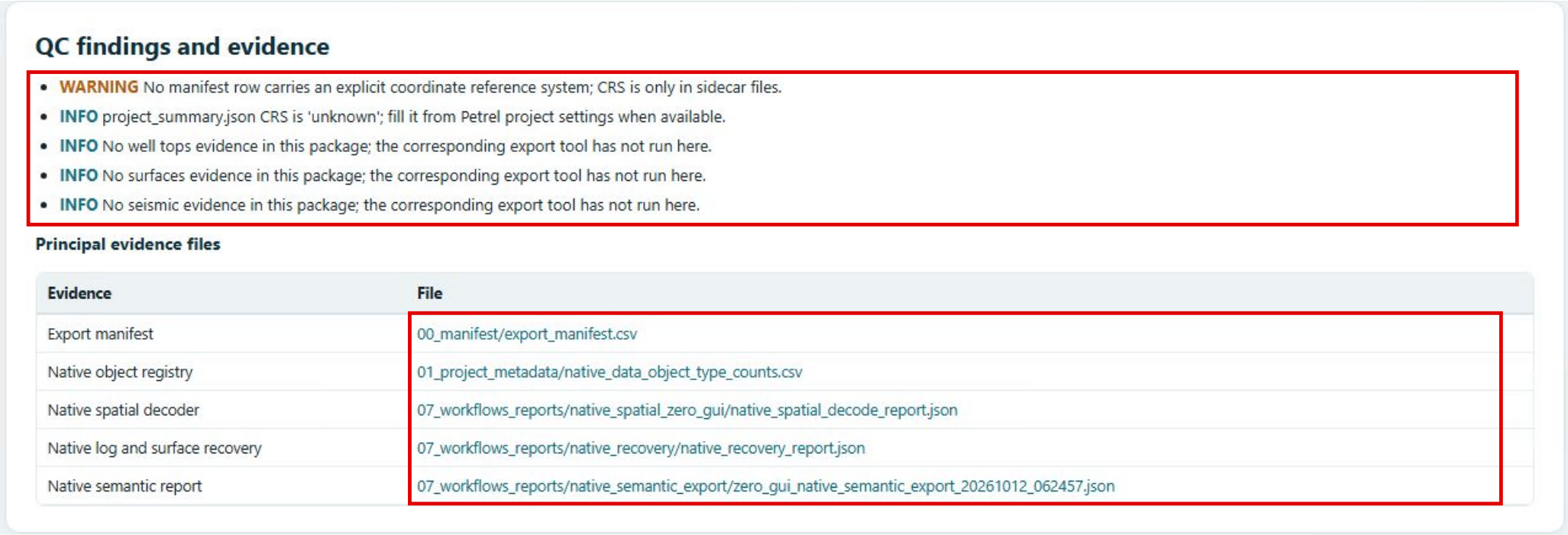
Read the final status, diagnostic link and category outcomes.

10. Check QC and metadata

1. Click **QC & evidence** in the report. Read the warnings and open the relevant evidence links.
2. Before importing a file, check its units, CRS and seismic axes in the matching **metadata.json**.



Click QC & evidence.



QC findings and evidence

- **WARNING** No manifest row carries an explicit coordinate reference system; CRS is only in sidecar files.
- **INFO** project_summary.json CRS is 'unknown'; fill it from Petrel project settings when available.
- **INFO** No well tops evidence in this package; the corresponding export tool has not run here.
- **INFO** No surfaces evidence in this package; the corresponding export tool has not run here.
- **INFO** No seismic evidence in this package; the corresponding export tool has not run here.

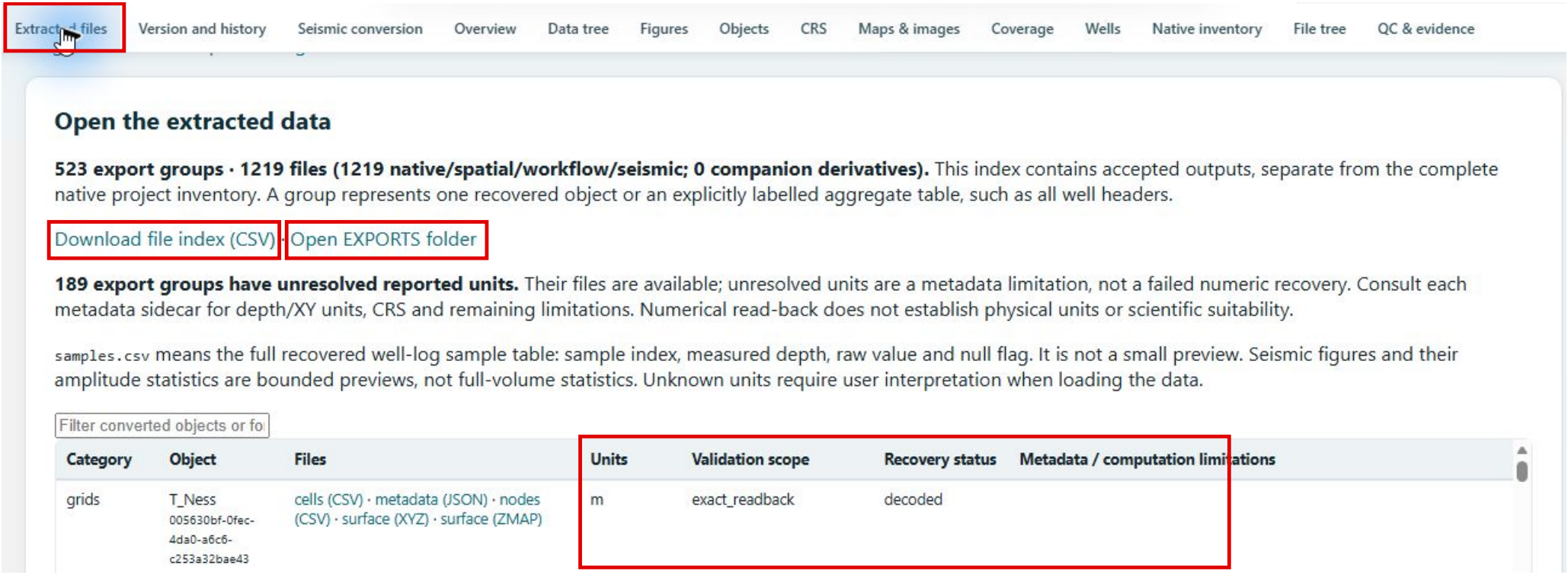
Principal evidence files

Evidence	File
Export manifest	00_manifest/export_manifest.csv
Native object registry	01_project_metadata/native_data_object_type_counts.csv
Native spatial decoder	07_workflows_reports/native_spatial_zero_gui/native_spatial_decode_report.json
Native log and surface recovery	07_workflows_reports/native_recovery/native_recovery_report.json
Native semantic report	07_workflows_reports/native_semantic_export/zero_gui_native_semantic_export_20261012_062457.json

Read the warnings and follow the relevant evidence links.

11. Open the extracted files

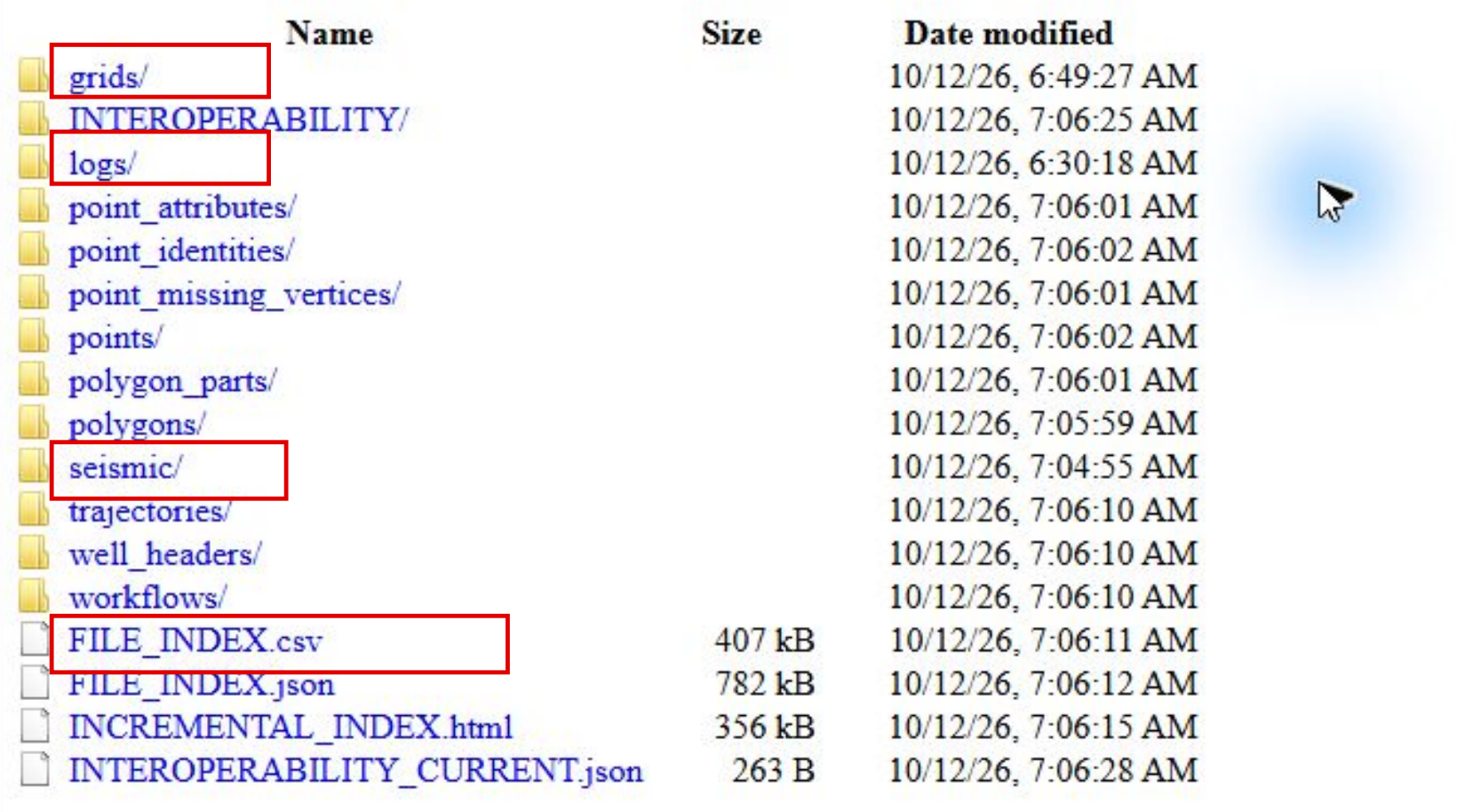
1. Click **Extracted files** in the report. Read the unit and recovery-status columns for the objects you need.
2. Click **Open EXPORTS folder** to browse the outputs. Use **Download file index (CSV)** to get the file list.
3. Keep data files with their metadata sidecars. Read unresolved-unit notices before using an export.



Use Extracted files, the CSV index and Open EXPORTS folder.

12. Find outputs by category

1. In **_EXPORTS**, open the category you need, such as **logs**, **grids** or **seismic**.
2. Open **FILE_INDEX.csv** for the full accepted file list. Filenames contain object identities to help match data to metadata.



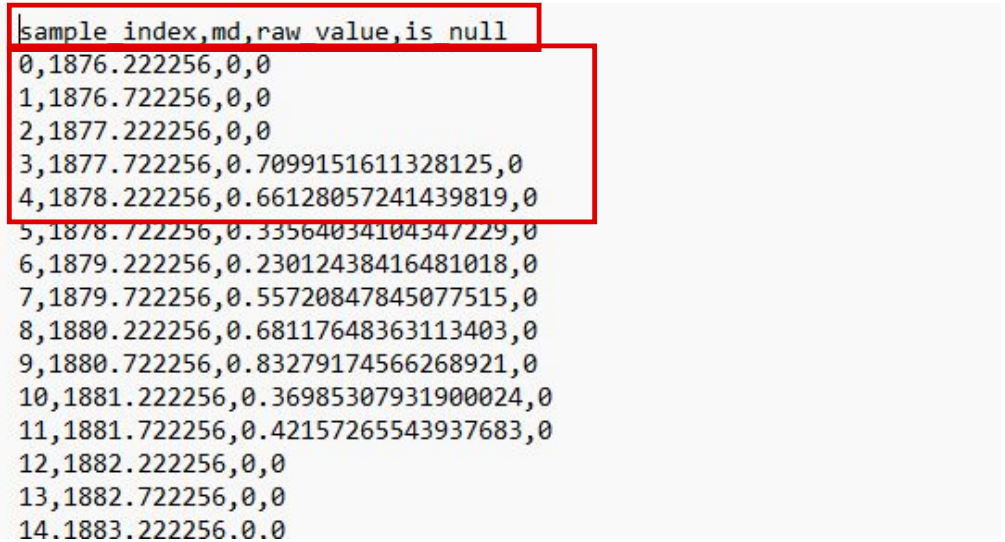
Name	Size	Date modified
grids/		10/12/26, 6:49:27 AM
INTEROPERABILITY/		10/12/26, 7:06:25 AM
logs/		10/12/26, 6:30:18 AM
point_attributes/		10/12/26, 7:06:01 AM
point_identities/		10/12/26, 7:06:02 AM
point_missing_vertices/		10/12/26, 7:06:01 AM
points/		10/12/26, 7:06:02 AM
polygon_parts/		10/12/26, 7:06:01 AM
polygons/		10/12/26, 7:05:59 AM
seismic/		10/12/26, 7:04:55 AM
trajectories/		10/12/26, 7:06:10 AM
well_headers/		10/12/26, 7:06:10 AM
workflows/		10/12/26, 7:06:10 AM
FILE_INDEX.csv	407 kB	10/12/26, 7:06:11 AM
FILE_INDEX.json	782 kB	10/12/26, 7:06:12 AM
INCREMENTAL_INDEX.html	356 kB	10/12/26, 7:06:15 AM
INTEROPERABILITY_CURRENT.json	263 B	10/12/26, 7:06:28 AM

Browse a category folder or open FILE_INDEX.csv.

13. Inspect data with its metadata

1. Open an exported **samples.csv** in a text editor or spreadsheet. Check the columns, raw values and null flags.

2. Open the matching **metadata.json**. Check **unit_status**, depth/axis units and **crs_status** for that object.



```
sample_index,md,raw_value,is_null
0,1876.222256,0,0
1,1876.722256,0,0
2,1877.222256,0,0
3,1877.722256,0.7099151611328125,0
4,1878.222256,0.66128057241439819,0
5,1878.722256,0.33564034104347229,0
6,1879.222256,0.23012438416481018,0
7,1879.722256,0.55720847845077515,0
8,1880.222256,0.68117648363113403,0
9,1880.722256,0.83279174566268921,0
10,1881.222256,0.36985307931900024,0
11,1881.722256,0.42157265543937683,0
12,1882.222256,0,0
13,1882.722256,0,0
14,1883.222256,0,0
```

Read the CSV column names and stored values.



```
"well": "C6",
"well_id": "bbaa2a07-0b4f-405e-bc2f-0d3a52a912b2",
"external_uwi": null,
"external_uwi_status": "unresolved; native well UUID is not an external UWI",
"template_id": "00000015-0000-0000-0000-000000000000",
"unit": "_",
"measurement": "Ratio",
"declared_template_unit": "_",
"unit_status": "validated_metric_profile",
"unit_resolution_gaps": [],
"depth_unit": "m",
"horizontal_unit": "m",
"project_units": {
  "xy": "m",
  "z": "m",
  "time": "ms",
  "system": "Metric"
},
"crs_status": "unresolved",
"source_sign_convention": "preserved; no time/depth sign reversal",
```

Read the matching unit fields and CRS status.

Red outline boxes show what to enter, click or read. Desktop controls and SLB 2018 demo outputs.

GitHub: https://github.com/ahmedsahernouh/petrel-headless-extractor